

Anomaly Detection for Real-World Industry Manufactured Products using Computer Vision

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Abstract— Human cognition naturally excels at identifying irregular patterns in images, enabling the distinction between expected variations and anomalies. Industrial defect classification presents unique challenges, encompassing a wide range of potential errors, from minor nuances to critical structural issues. This research work introduces PaDiM (Patch Distribution Modeling Framework) model, a novel approach for cold-start anomaly detection in industrial imagery. Prior research predominantly focuses on acquiring a model representing the standard distribution, often achieved through techniques like auto encoding, GANs, or other unsupervised adaptation methods. Even without explicit adaptation, these models exhibit robust anomaly detection capabilities, effectively localizing defects within the spatial context. PaDiM excels in industrial anomaly localization, positioning itself as a state-of-the-art solution. It maintains efficiency with swift inference times, obviating the need for specific dataset training. This characteristic renders PaDiM highly attractive for practical applications in industrial anomaly detection.

In this research work, PaDiM model is implemented for real-world industry manufactured products such as Bottle, Cable, Capsule, Hazelnut and Screw using Computer Vision for anomaly detection on MVTec AD dataset. And obtained an accuracy of 83%. Additional experiments highlight PaDiM high sample efficiency, matching the performance of existing anomaly detection methods while utilizing only a fraction of the nominal training data. Furthermore, PaDiM's versatility extends to its adaptability in various industrial settings. Its ability to effectively identify and classify a wide range of defects, from subtle imperfections to major structural discrepancies, showcases its potential for widespread applicability across diverse manufacturing processes. This adaptability is a testament to Patch Core's robustness and underscores its significance as a powerful tool in modern industrial quality control efforts.

Keywords—Industrial Anomaly Detection, Generative adversarial networks (GANs), Artificial Intelligence(AI), Deep Learning(DL), PaDiM, Computer Vision(CV), Machine Learning.

I. INTRODUCTION

Human cognition is naturally equipped with the capability to identify irregular patterns in images. Even with exposure to just a limited set of typical cases, humans can distinguish between anticipated fluctuations in the data and anomalies. Classifying defects in industrial visuals poses a particular challenge due to the wide range of potential errors,

spanning from minor nuances like thin scratches to more significant structural issues such as missing components. Prior research on cold-start anomaly detection in industrial visuals predominantly centers on the acquisition of a model representing the standard distribution, typically accomplished through techniques like autoencoding, GANs, or other unsupervised adaptation methods. Comprehensive experimentation carried out on both the diverse MVTec AD dataset and the specialized Magnetic Tile Defects (MTD) industrial anomaly detection benchmarks highlights the effectiveness of PaDiM in industrial anomaly detection. PaDiM sets new standards in image-level detection scores for both MVTec AD and MTD, achieving nearly flawless performance on MVTec AD (with an AUROC of up to 99.6%), thereby reducing detection errors by over half when compared to previous methods. It also excels in industrial anomaly localization, establishing itself as a state-of-the-art solution. Importantly, PaDiM maintains its efficiency by providing swift inference times without necessitating training on the specific dataset. This characteristic renders PaDiM exceptionally appealing for practical applications in industrial anomaly detection [17][18][19]. Furthermore, additional experiments underscore PaDiM's high sample efficiency, matching the performance of existing anomaly detection methods while utilizing only a fraction of the nominal training data.

II. LITERATURE SURVEY

The literature review constitutes a pivotal element of this research, offering a thorough examination of prior studies and their discoveries within the field. Its purpose is to amalgamate and evaluate the pertinent body of knowledge, pinpointing areas where this study offers novel insights and addresses gaps in existing research. This section forms the bedrock upon which this investigation is constructed, providing a comprehensive understanding of the present state of the field and guiding the research's objectives and methodology.

K. Roth *et al* [1] described identifying faulty components in large-scale industrial production, particularly focusing on the challenging cold-start problem. The goal is to create automated systems capable of efficiently handling diverse tasks without relying on manual solutions for individual classes. Y. Jiang *et al* [2] addresses the crucial task of anomaly detection in industrial production, crucial

for ensuring product quality. Deep learning techniques are employed for this purpose. For balanced datasets, a supervised model based on YOLOv3 is introduced, incorporating ROI classifier to identify various anomalies. In cases of unbalanced datasets, a semi-supervised model using Fast-AnoGAN is presented, leveraging normal samples and a pre-trained WGAN-GP model for anomaly detection. Evaluation on real-world industrial datasets confirms the models' effectiveness in real-time, high-accuracy anomaly detection in dynamic production environments. J. Xu *et al* [3] discuss the rising concern of information technology posing risks to secure industrial control systems amid increasing integration. Identifying anomalous network behaviour becomes a crucial challenge in industrial control security. The paper introduces an ensemble learning method with feature fusion to address anomaly classification in network data. Computational experiments confirm the method's effectiveness, showcasing its proficiency in detecting diverse network attacks and enhancing overall accuracy. This research presents a promising approach to strengthen the security of industrial control systems in evolving technological environments. W. Shen *et al* [4] emphasizes the pivotal role of anomaly detection in securing devices like sensors and actuators in Industrial Control Systems (ICS). Traditional deep learning-based methods forecast or reconstruct device states, coupled with an anomaly determination mechanism. However, cyberattacks in ICSs often manifest as continuous anomaly segments, challenging the fixed threshold approach. To address this, a novel detection mechanism is introduced, focused on identifying the onset and conclusion times of anomalies based on their consistent characteristics and error dynamics. Experimental results on authentic datasets exhibit a significant improvement in the F1 score, indicating a notable enhancement in real-time anomaly detection performance. W. Wang *et al* [5] highlights the critical importance of anomaly detection in Industrial Control Systems (ICSs), which form the backbone of a nation's critical infrastructure. To address challenges related to gradient instability and prolonged training times, a novel model based on a deep residual CNN is introduced. The methodology employs transfer learning and modifies the CNN structure to detect both known and unknown attacks. 1D ICS flow data is transformed into 2D grayscale images to leverage CNN's feature extraction capabilities. The results demonstrate the effectiveness of this approach, providing reliable predictions for abnormal data, thereby enhancing the safety of ICSs.

J. Zou *et al* [6] highlights the significance of Industrial Control Security (ICS) in protecting industrial assets from potential malicious interference. It addresses the lack of real-world case studies validating ICS approaches in threat mitigation. The paper presents a tangible case study conducted in industrial settings, examining the spread of viruses and propagation of worms. It demonstrates the deployment of anomaly detection techniques to identify malicious behaviours, providing crucial support to security administrators in enhancing industrial security. This case study vividly showcases the pivotal role of anomaly detection in bolstering the resilience of ICS against evolving threats. B. Siegel *et al* [7] emphasizes the crucial role of anomaly detection in ensuring the secure operation of

industrial equipment and integrated sensors. In cyber-physical systems, algorithms are integrated for effective anomaly detection, alleviating the labour-intensive manual identification process due to the rarity of anomalies. Experimental results on substantial multivariate time-series datasets demonstrate the superiority of these neural network architectures over conventional machine learning techniques in terms of detection rates. Particularly, the IDCNN proves to be a valuable alternative to RNNs, especially for handling long time-sequences. This research highlights the effectiveness of deep learning techniques in enhancing anomaly detection capabilities in industrial contexts. W. Shang *et al* [8] addresses the threat of viruses and Trojans targeting the application layer network protocol in industrial control systems. The study proposes an intrusion detection approach utilizing a combination of unsupervised fuzzy C-means clustering (FCM) and SVM techniques. This method calculates the distance between communication data and the cluster centre, followed by classification using SVM for partial data meeting a threshold condition. Experimental results demonstrate the method's effectiveness, significantly reducing training time and improving classification accuracy compared to traditional techniques, without requiring prior knowledge of class labels. This research represents a notable advancement in enhancing the security of industrial control systems against malicious attacks on the application layer network protocol. W. Wang *et al* [9] This paper addresses the vulnerability of power industrial control systems due to the lack of initial security considerations in their communication protocols. Anomaly detection within these protocols is crucial for strengthening their defence capabilities against cyber threats. The paper introduces an approach using Recurrent Neural Networks (RNNs) for detecting anomalies in unknown protocols within the power industrial control system. Through simulations with real substation traffic data, the method proves highly effective, showcasing high accuracy and low false alarm rates, surpassing traditional machine algorithms in this context. H. Yan *et al* [10] introduces an Adaptive Multi-layer Structure for hyperspectral image anomaly detection, addressing the neglect of local spatial structure in traditional algorithms. The approach evaluates both spectral differences and spatial structure distinctions between target and background pixels. The comprehensive framework includes anomaly detection, background suppression, and criteria layers, outperforming conventional models by effectively incorporating spatial and spectral information, reducing false alarms, and improving abnormal target detection with limited pixel data.

S. Akcay *et al* [11] introduces anomalib, an open-source library focused on unsupervised anomaly detection and localization. It emphasizes reproducibility and modularity, offering established algorithms and tools for creating custom anomaly detection methods. anomalib includes cutting-edge techniques with top-tier performance, along with resources for customization and additional tools for experiment tracking and model optimization, making it a comprehensive resource for conceiving, implementing, and deploying unsupervised anomaly detection models. H. Zhu *et al* [12] described timely detection of appearance defects in notebooks is crucial for quality control in industrial production. However, challenges persist due to the rarity of

defect samples and subtle differences in high-resolution images. The proposed method combines unsupervised Patch Core training on normal samples with Defect GAN to generate realistic abnormal samples, achieving an impressive AUROC score of 96.1% for image-level anomaly detection on TKP-Surface datasets. This approach overcomes limitations in current methods, providing a robust framework for notebook appearance defect detection in industrial applications. K. Ishida *et al* [13] SA-Patch Core introduces self-attention to enhance local anomaly detection, addressing the challenge of co-occurrence-related anomalies often absent in existing datasets. The newly developed CAD-SD dataset demonstrates that SA-Patch Core outperforms Patch Core in detecting co-occurrence-related anomalies, while maintaining comparable performance in the MVTec Anomaly Detection dataset, which focuses on anomalies in local areas. T. Ma *et al* [14] introduces the Orthogonal Features Fusion Network (OFF-Net) as a novel architecture to improve generative models' efficiency in anomaly detection. OFF-Net emphasizes the orthogonality of convolutional feature maps, enhancing representation capacity and temporal connections between consecutive images. It can be seamlessly integrated into convolutional neural networks (CNNs) with minimal additional parameters, outperforming state-of-the-art models by 1.7% in AUC, while also significantly reducing parameter count compared to prevailing methods using optical flow networks. U. Rehman *et al* [15] This research tackles anomaly detection in outdoor surveillance using a novel approach that combines visual and audio data. It employs visual elements like optical flow optimization and the social force model, alongside acoustic features including energy, zero crossing rate, and spectral attributes. The method incorporates both modalities into an anomaly inference mechanism, outperforming existing methods in accuracy when tested on UMN datasets with audio recordings. Y. Xu *et al* [16] addresses the identification of leaked heat pipelines using aerial thermal imagery, aiming to reduce false alarms common in current detection methods. Leveraging the human visual system, it employs a saliency map from thermal images to spot potential leaks. A subsequent buffer analysis with a pipeline GIS layer helps eliminate erroneous detections, resulting in superior detection rates, particularly in scenarios with limited prior knowledge, and demonstrating increased versatility in real-world conditions.

Collectively, the reviewed literature emphasizes the vital role of advanced anomaly detection techniques in various industrial domains. Innovative approaches like Patch Core, SA-Patch Core, and OFF-Net showcase significant progress in detecting anomalies, even with limited anomaly data. The introduction of novel datasets and methodologies, such as CAD-SD and OFF-Net, demonstrates a marked improvement in overcoming existing limitations, promising more effective anomaly detection systems for diverse industrial applications.

III. DESIGN AND IMPLEMENTATION

The anomaly detection process is achieved through the implementation of the PaDiM (Patch Distribution Modeling Framework) model. PaDiM utilizes a novel approach for

cold-start anomaly detection in industrial imagery. The model is trained on the MVTec AD dataset, and it excels in industrial anomaly localization by effectively capturing the spatial context of anomalies without the need for specific dataset training. The process involves training the model on normal and defective images, and during testing, the model identifies anomalies by comparing the input image against the learned distribution of normal patches.

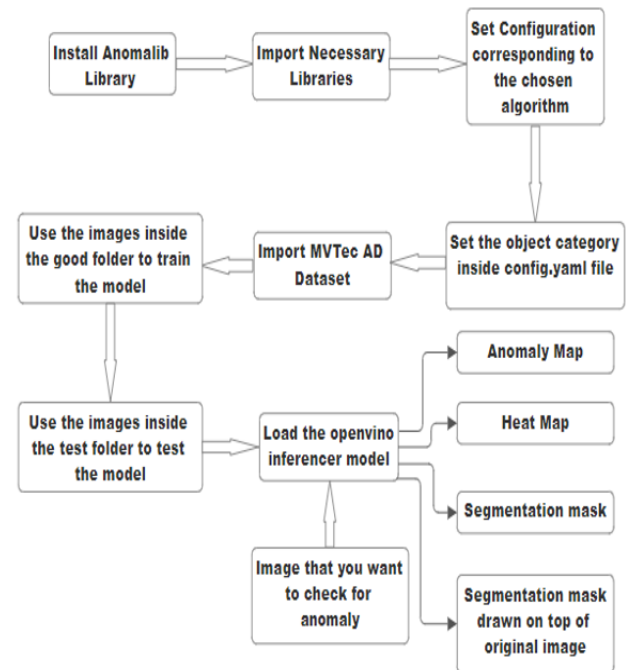


Fig. 1. Block diagram of design and implementation

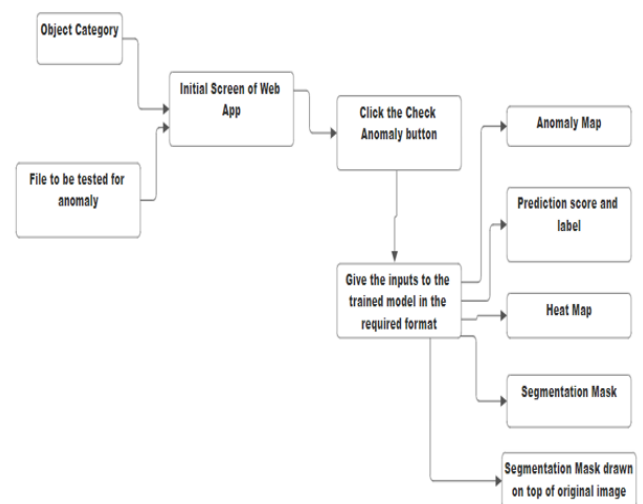


Fig. 2. Data flow diagram of implementation

Fig.1 and Fig.2. shows the block diagram and dataflow diagram of design and implementation. The Methodology followed in the industrial anomaly detection as follows.

Installing anomalib-anomalib is a deep learning library that aims to collect state-of-the-art anomaly detection algorithms for benchmarking on both public and private datasets.

Importing Libraries: Libraries imported are NumPy, matplotlib, PIL, pytorch_lightning, torchvision.transforms.

Setting Configuration- We use the default config file that comes with Padim implementation, which uses ./datasets/MVTec as the path to the dataset. We overwrite this after loading the config.

Importing Dataset-MVTec AD is a dataset for benchmarking anomaly detection methods with a focus on industrial inspection. It contains over 5000 high-resolution images divided into 15 different object and texture categories. Each category comprises a set of defect-free training images and a test set of images with various kinds of defects as well as images without defects. If the dataset is not located in the root datasets directory, anomalib will automatically install the dataset.

Visualizing an image with a mask-Visualize a normal and abnormal sample from the validation set.

Splitting the Dataset-The dataset is split into training data and testing data.

Training the Model-The PaDiM Model is trained using the training image dataset.

Testing the Model-The PaDiM Model is tested using the testing dataset and accuracy of the model is found.

Checking OpenVINO Inference results-Once the testing is done the results are stored in results folder which contains for each image the anomaly map, heat map, prediction mask and superposition of original image and mask.

Providing Custom Input-One of the test images is fed into the model, and the resulting output is presented on the web interface which shows if anomaly is present or not.

IV. RESULTS AND ANALYSIS

This section describes the detailed results and analysis of 5 categories of products namely Bottle, Cable, Capsule, Hazelnut and Screw. Confusion matrix is plotted considering all five 5 categories of products. This analysis provides useful insights to better understanding. For each of these 5 categories, defect types and good images have been shown.

The dataset used for this research is the MVTec AD dataset, which is a comprehensive real-world dataset designed for unsupervised anomaly detection. It consists of over 5,000 high-resolution images categorized into 15 different object and texture categories. The categories include products such as Bottle, Cable, Capsule, Hazelnut, and Screw. Each category comprises defect-free training images and a test set with various defects. The MVTec AD dataset is widely recognized in the anomaly detection community.

The prediction score is calculated based on the model's output during testing. Specifically, after feeding an image to the trained PaDiM model, the model produces an anomaly map, which highlights regions in the image where anomalies are detected. The prediction score is obtained by evaluating the extent and intensity of anomalies in this map. The higher the score, the more likely the presence of anomalies. The details of the scoring mechanism can be found in the PaDiM framework, particularly in the prediction mask generation process.

The segmentation task involves the generation of a predicted mask that indicates where the model identifies anomalies in the input image. This task is performed by thresholding the anomaly map generated by the PaDiM model. Pixels exceeding a certain threshold are classified as

anomalies, contributing to the final segmentation mask. The segmentation mask provides a clear visual representation of areas in the image where the model predicts anomalies.

The efficiency of the anomaly detection process is maintained through several strategies. First, the PaDiM model is designed for cold-start anomaly detection, eliminating the need for specific dataset training. This enhances efficiency by enabling swift inference times during testing. Additionally, the use of PaDiM showcases high sample efficiency, achieving comparable performance to existing anomaly detection methods while utilizing only a fraction of the nominal training data. These features contribute to the overall efficiency of PaDiM in practical industrial applications.

Case 1: Broken Bottle and Contaminated Bottle

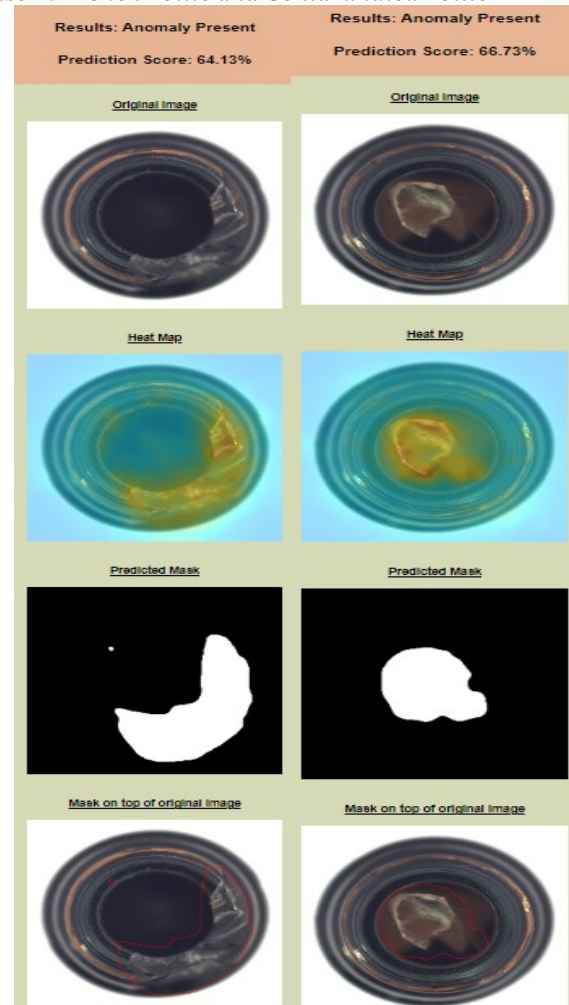


Fig 3. Defect Type- Broken Bottle and Contaminated Bottle

Case 2: Good Bottle and Cable with bent wire

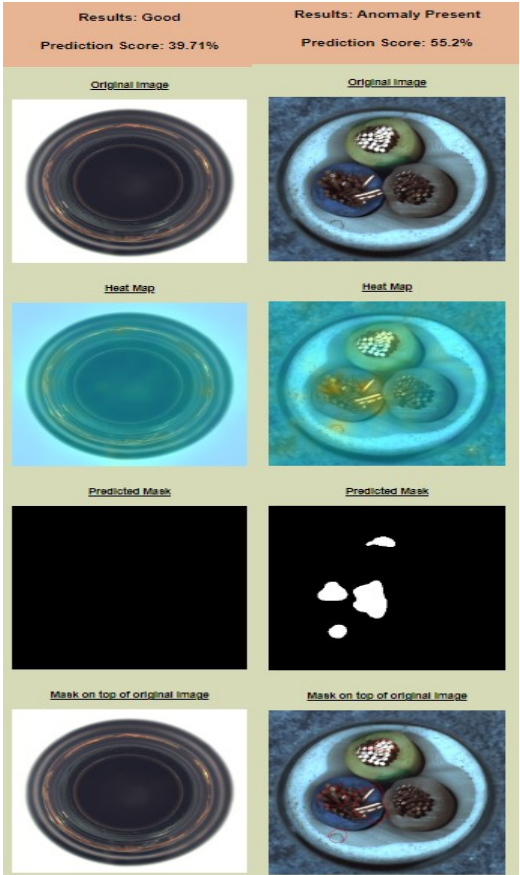


Fig 4. Defect Type-Good Bottle and Cable with bent wire
Case 3: Cable with Inner Insulation cut and Good Cable

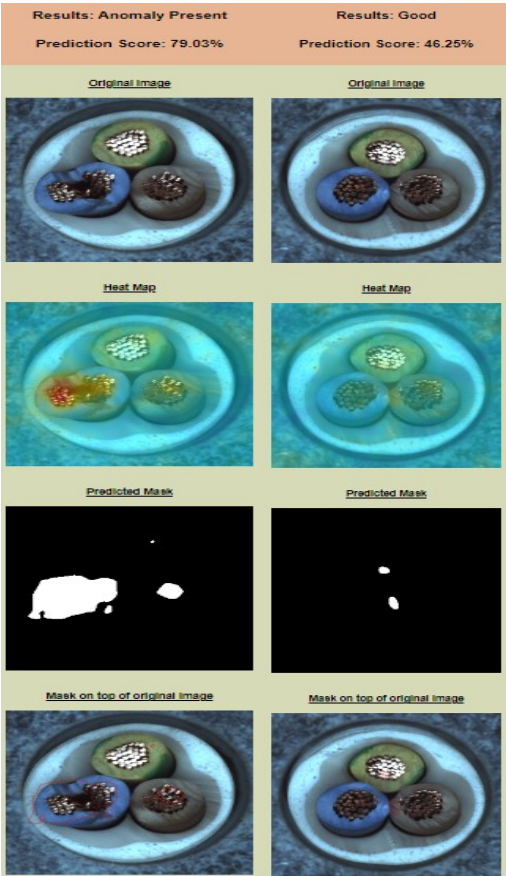


Fig.5 Defect Type-Cable with Inner Insulation cut and Good Cable

Case 4: Capsule with crack and Squeezed Capsule

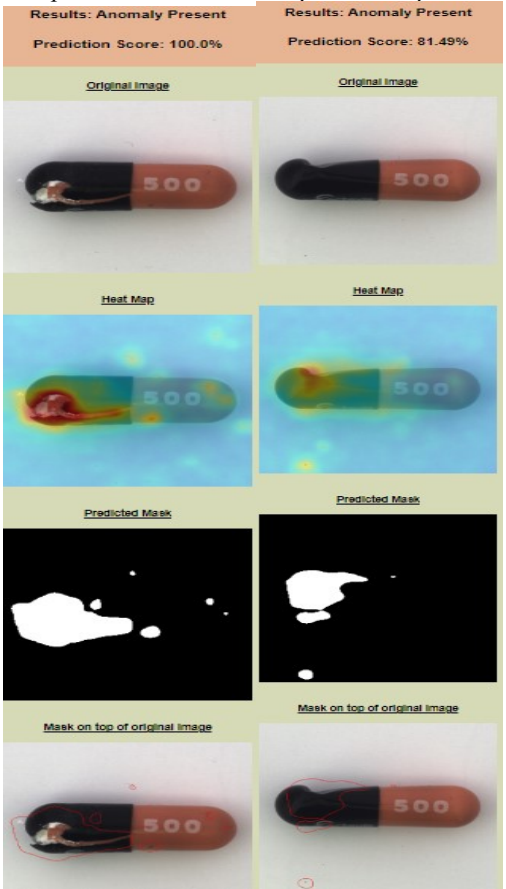


Fig.6. Defect Type- Capsule with crack and Squeezed Capsule
Case 5: Good Capsule and Cracked Hazelnut

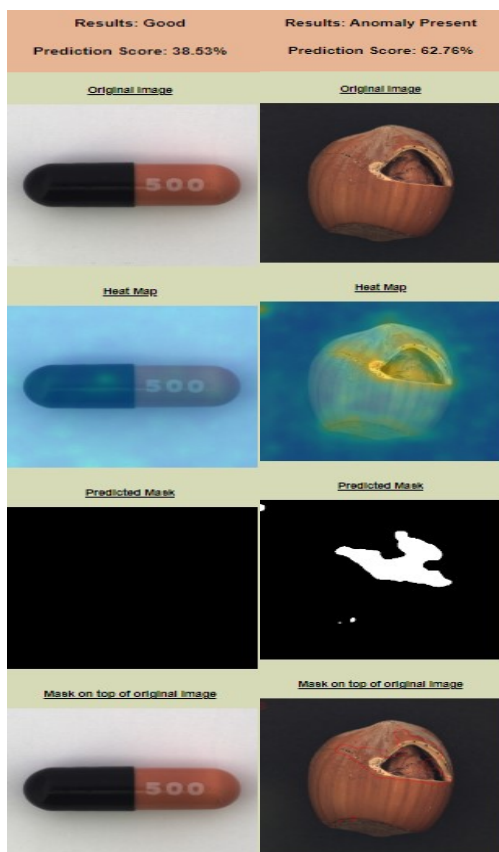


Fig.7. Defect Type- Good Capsule and Cracked Hazelnut

Case 6: Hazelnut with a hole and Good Hazelnut

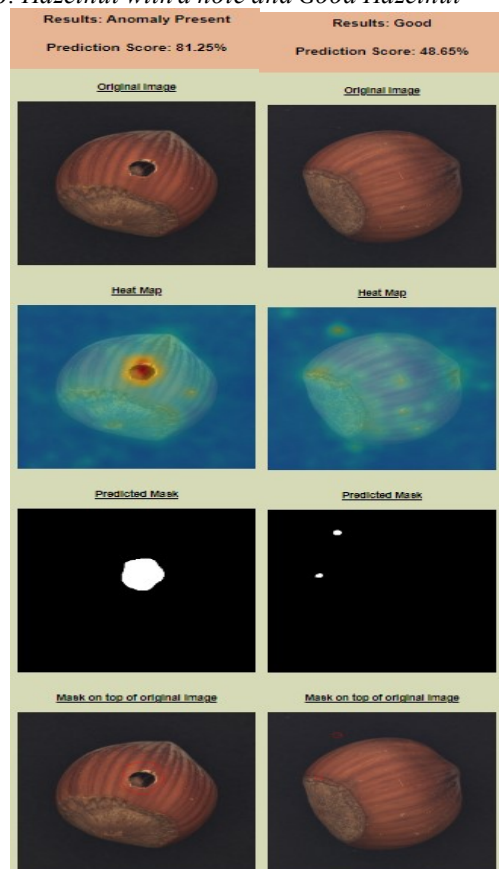


Fig.8. Defect Type- Hazelnut with a hole and Good Hazelnut

Case 7: Screw with a manipulated front, Screw with a scratched head and Good Screw

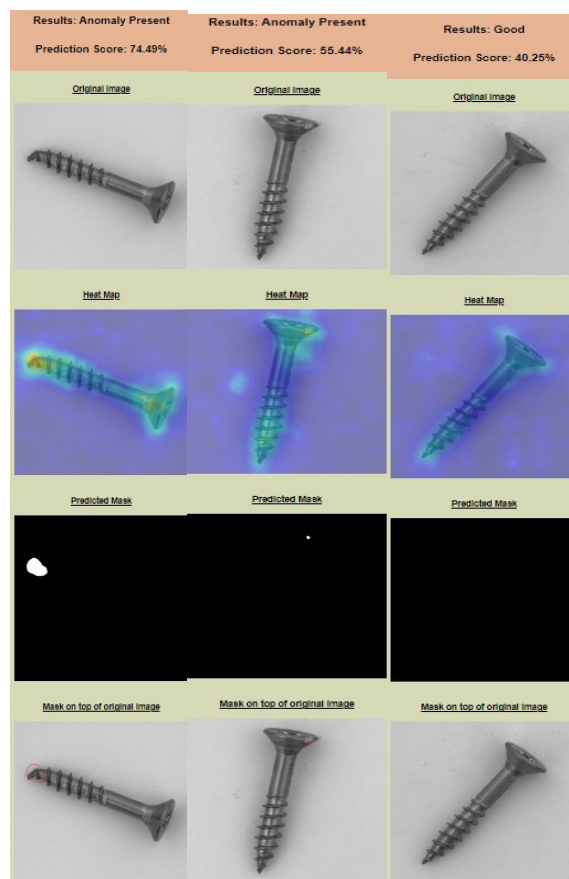


Fig.9. Defect Type- Screw with a manipulated front, Screw with a scratched head and Good Screw

Figure 3 to 7 shows the results of various defective products and good products with prediction score. In all figures the first image shows the original image which was given as input to the model. The second image shows the heat map. The third image shows the predicted mask indicating where the model thinks anomaly is present. The fourth image shows the mask drawn on top of the original image.

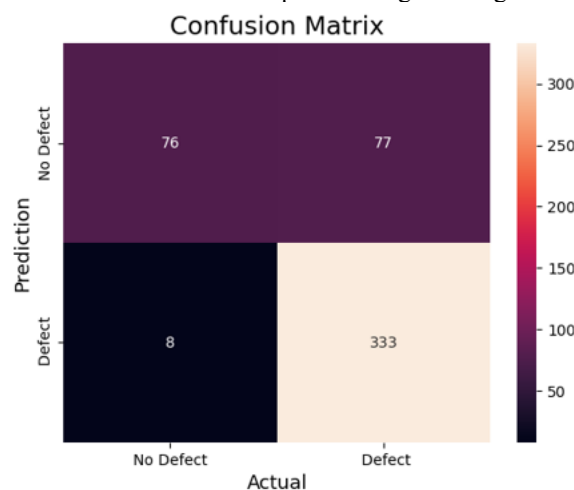


Fig.10. Confusion Matrix

Figure 10 shows the confusion matrix. 76 images in the dataset for which the ground truth is no defect, and the model output is also no defect. It also tells that there are 353 images for which ground truth is defect and the model output is also defect. The Confusion matrix further reveals that there are 77 images for which ground truth is defect, but model output is no defect. It also shows that there are 8

images for which ground truth is No Defect, but model output is defect. So, the obtained accuracy of the model is 83%.

High accuracy is achieved through the robustness of the PaDiM model in capturing and localizing anomalies within industrial imagery. The model's ability to represent the standard distribution without explicit adaptation, coupled with its effective anomaly localization, contributes to its high accuracy. The model is trained on diverse categories of products, and its performance is evaluated on the MVTec AD dataset, achieving an accuracy of 83% for products such as Bottle, Cable, Capsule, Hazelnut, and Screw.

V. LIMITATIONS AND CHALLENGES OF REAL-TIME IMPLEMENTATION

Acknowledging the limitations of existing work is crucial for providing a comprehensive view of the research landscape. In the literature review, it's important to note that while innovative approaches like PaDiM, Patch Core, SA-Patch Core, and OFF-Net showcase significant progress in detecting anomalies, challenges such as limited anomaly data persist. Additionally, the diversity of industrial settings and the variability in anomaly types pose ongoing challenges for anomaly detection methods. By acknowledging these limitations, the paper emphasizes the need for continuous improvement and adaptation in the field of industrial anomaly detection.

The proposed study is positioned for significant contributions to both the future scope of research and real-time implementation. In terms of research, our novel approach, PaDiM (Patch Distribution Modeling Framework), introduces a robust methodology for cold-start anomaly detection in industrial imagery. This framework not only addresses current challenges in anomaly detection but also lays the foundation for further advancements in the field.

Furthermore, the versatility of PaDiM is a key factor in its suitability for real-time implementation. Its efficiency, as demonstrated by swift inference times and high accuracy, makes it practical for deployment in industrial settings where timely anomaly detection is crucial. The absence of specific dataset training requirements enhances its adaptability to diverse manufacturing processes, emphasizing its potential for widespread application in real-world scenarios.

VI. CONCLUSION

In this research work, PaDiM (Patch Distribution Modeling Framework) model is implemented for anomaly detection on the MVTec AD dataset. Remarkably, achieved an accuracy of 83% for five category of products such as Bottle, Cable, Capsule, Hazelnut and Screw. To enhance applicability, meticulously saved five distinct models, each tailored to specific object categories: Bottle, Cable, Capsule, Hazelnut, and Screw. Further to provide a user-friendly interface, developed a web application using Flask. This application streamlines the testing process by prompting users to input the object category and upload the image they wish to assess for anomalies. The result page displays vital visualizations, including the anomaly map, segmentation mask, and the segmentation mask

superimposed onto the original image. This implementation not only showcases the successful integration of the PaDiM model but also demonstrates the effectiveness of deploying Flask for web application development. This user-centric approach significantly simplifies the anomaly detection process for end-users, making it accessible and straightforward.

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